## PCM REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

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**Note**
- If the PCM is replaced, perform the PCM Replacement procedure. (See PCM Replacement.)

### PCM Replacement

**Note**
- The PCM replacement differs depending on whether As-built data is used or not during the PCM configuration.

**Replacement procedure determination**

| Step | Action | Results | Action |
|---|---|---|---|
| 1 | Perform the PCM configuration and verify if the As-Built data input screen is displayed or not.<br>(See PCM CONFIGURATION (USING READ/WRITE FUNCTION) [SKYACTIV-D 1.5].) | Not displayed | Go to “Procedure for PCM replacement (pattern 1)”.<br>(See Procedure for PCM replacement (pattern 1).) |
| | | Displayed | Go to “Procedure for PCM replacement (pattern 2)”.<br>(See Procedure for PCM replacement (pattern 2).) |

**Procedure for PCM replacement (pattern 1)**

| Step | Action | Page |
|---|---|---|
| 1 | Continue to perform the configuration. | — |
| 2 | Clear the DTCs. | (See CLEARING DTC [SKYACTIV-D 1.5].) |
| 3 | Wait for 20 s or more. | — |
| 4 | Perform the immobilizer system-related part programming. | (See IMMOBILIZER SYSTEM-RELATED PARTS PROGRAMMING.) |
| 5 | Perform the KOEO self test. | (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) |
| 6 | Perform the KOER self test. | (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) |
| 7 | Perform the Battery Condition Initialization Setting (i-stop setting) Procedure. | (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).) |
| 8 | Perform the "FUEL INJECTOR INJECTION AMOUNT CORRECTION" procedure. | (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) |

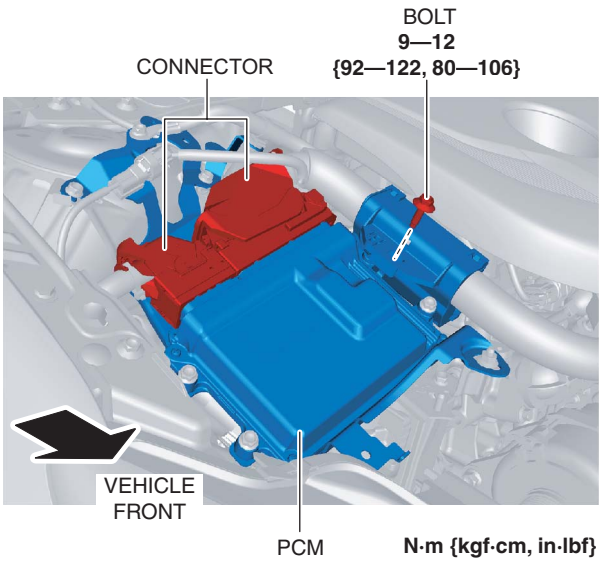
**Procedure for PCM replacement (pattern 2)**

| Step | Action | Page |
|---|---|---|
| 1 | Perform the configuration using the As-Built data. | (See PCM CONFIGURATION (USING AS-BUILT DATA) [SKYACTIV-D 1.5].) |
| 2 | Store the identification number of the fuel injector equipped to the vehicle. | — |
| 3 | Perform the fuel injector code programming. | (See FUEL INJECTOR CODE PROGRAM [SKYACTIV-D 1.5].) |
| 4 | Perform the PCM data reset procedure. | (See PCM DATA RESET [SKYACTIV-D 1.5].) |
| 5 | Perform the immobilizer system-related part programming. | (See IMMOBILIZER SYSTEM-RELATED PARTS PROGRAMMING.) |
| 6 | Clear the DTCs. | (See CLEARING DTC [SKYACTIV-D 1.5].) |
| 7 | Wait for 20 s or more. | — |
| 8 | Perform the KOEO self test. | (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) |
| 9 | Perform the KOER self test. | (See KOEO/KOER SELF TEST [SKYACTIV-D 1.5].) |
| 10 | Perform the Battery Condition Initialization Setting (i-stop setting) Procedure. | (See BATTERY CONDITION INITIALIZATION SETTING (i-stop SETTING).) |
| 11 | Perform the “FUEL INJECTOR INJECTION AMOUNT CORRECTION” procedure. | (See FUEL INJECTOR INJECTION AMOUNT CORRECTION [SKYACTIV-D 1.5].) |
| 12 | Perform compulsory DPF regeneration. | (See COMPULSORY DIESEL PARTICULATE FILTER REGENERATION [SKYACTIV-D 1.5].) |

| Step | Action | Page |
|---|---|---|
| 13 | Perform the following procedure:<br>1. Warm up the engine until the engine coolant temperature is 70 °C {158 °F} or more.<br>2. Switch the ignition off.<br>3. Store “Complete (delete record)” the M-MDS session.<br>4. Remove the M-MDS.<br>5. Start the engine.<br>6. Leave for 3 min while idling.<br>7. Continuously drive the vehicle for 30 s at a vehicle speed of 50 km/h {31 mph} or more.<br>— When the warning light turns on or flashes, repeat from step 5 (start the engine).<br>8. After stopping the vehicle, leave it idling for 3 min without stopping the engine.<br>9. Continuously drive the vehicle for 30 s at a vehicle speed of 50 km/h {31 mph} or more again.<br>— When the warning light turns on or flashes, repeat from step 5 (start the engine). | — |
| 14 | Verify that the learning value (O2S11_CAL) for the A/F sensor is other than 0.<br>— If the learning value (O2S11_CAL) for the A/F sensor is 0, repeat from step 5 of the procedure in Step 13 (start the engine). | (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) |
| 15 | Clear the DTCs. | (See CLEARING DTC [SKYACTIV-D 1.5].) |
| 16 | Replace the engine oil.<br>**Note**<br>• Because the PCM is new, it is not necessary to perform the engine oil data reset procedure. | (See ENGINE OIL REPLACEMENT [SKYACTIV-D 1.5].) |

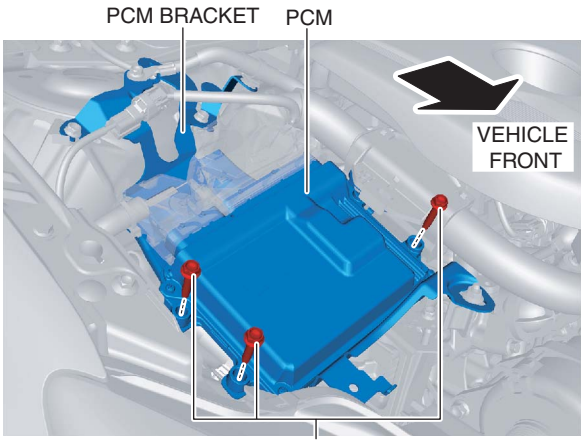
**PCM Removal/Installation**
**Without set bolt**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/CONNECTION.)
2. Remove the bolt.
3. Disconnect the PCM connectors. (See PCM Connector Connection Note.)



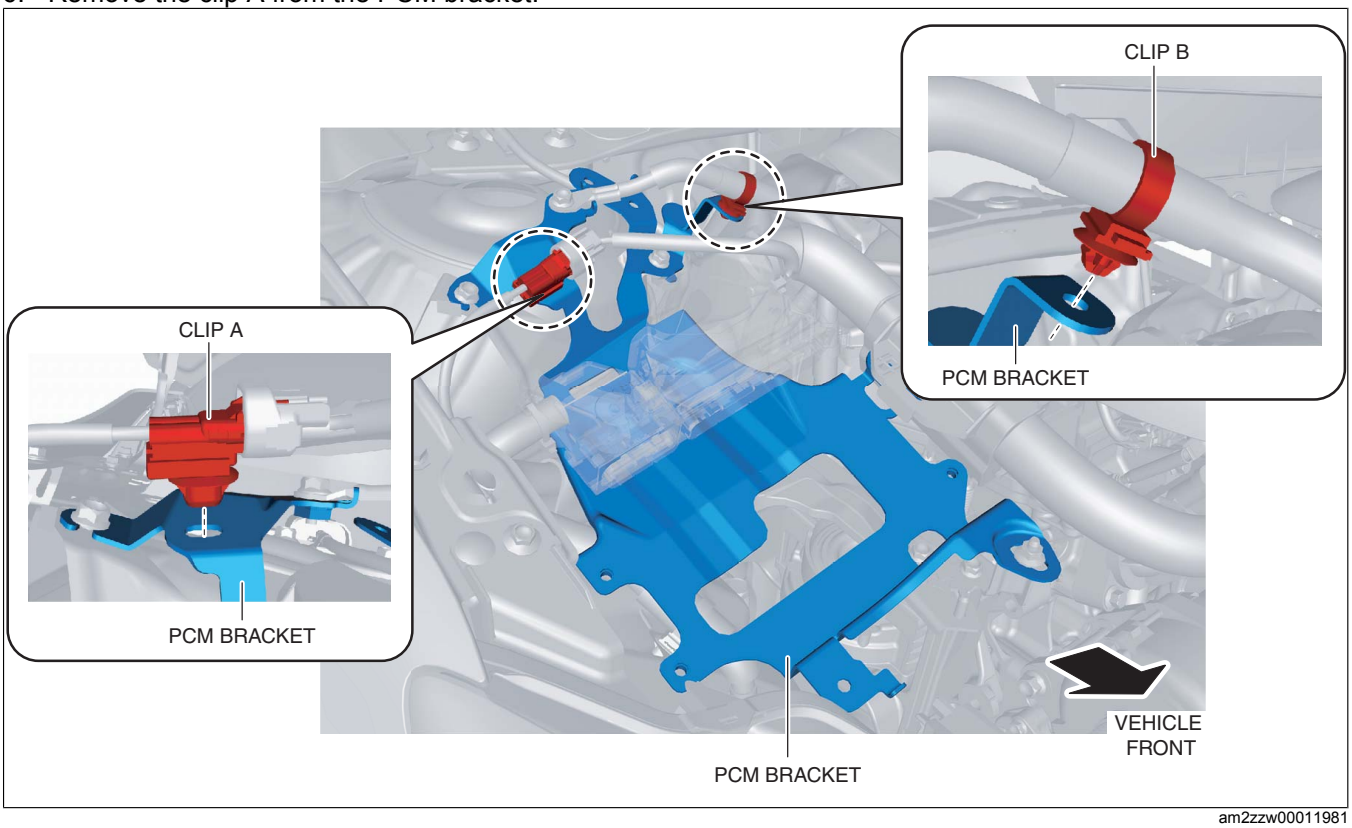

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4. Remove the bolt.
5. Remove the PCM.




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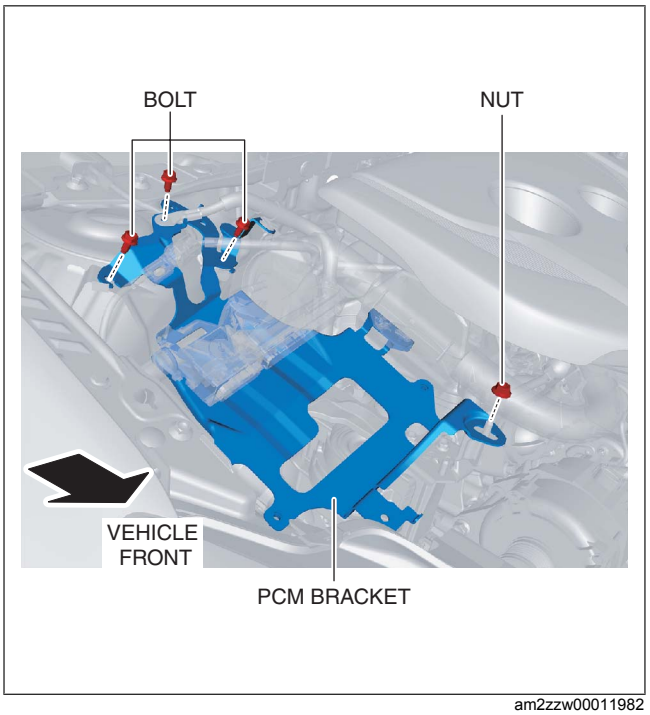
6. Remove the clip A from the PCM bracket.




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7. Remove the clip B from the PCM bracket.

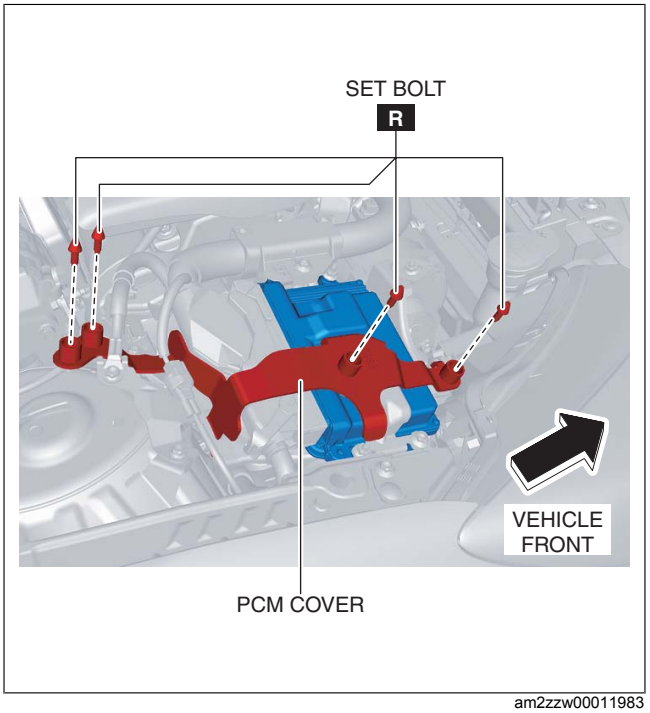
8. Remove the nut.
9. Remove the bolt.
10. Remove the PCM bracket. (See PCM Bracket Installation Note.)
11. Install in the reverse order of removal.
12. When replacing the PCM on the vehicles, perform the “PCM Replacement”. (See PCM Replacement.)



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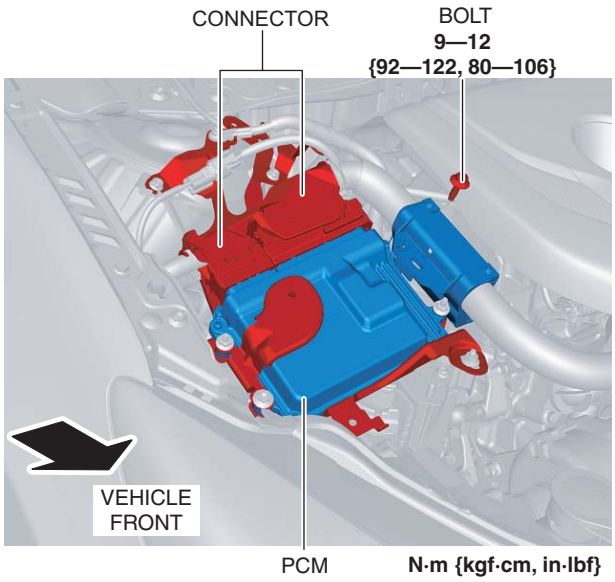
**With set bolt**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the PCM cover. (See Set Bolt Removal Note.) (See Set Bolt Installation Note.) (See PCM Cover Installation Note.)



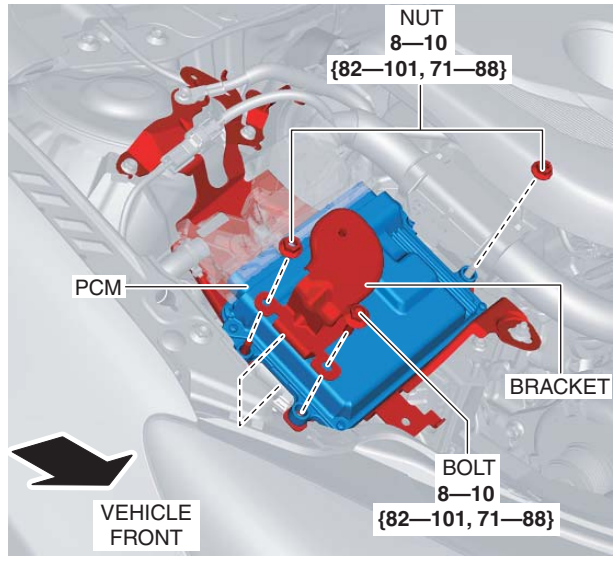
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3. Remove the bolt.
4. Disconnect the PCM connectors. (See PCM Connector Connection Note.)



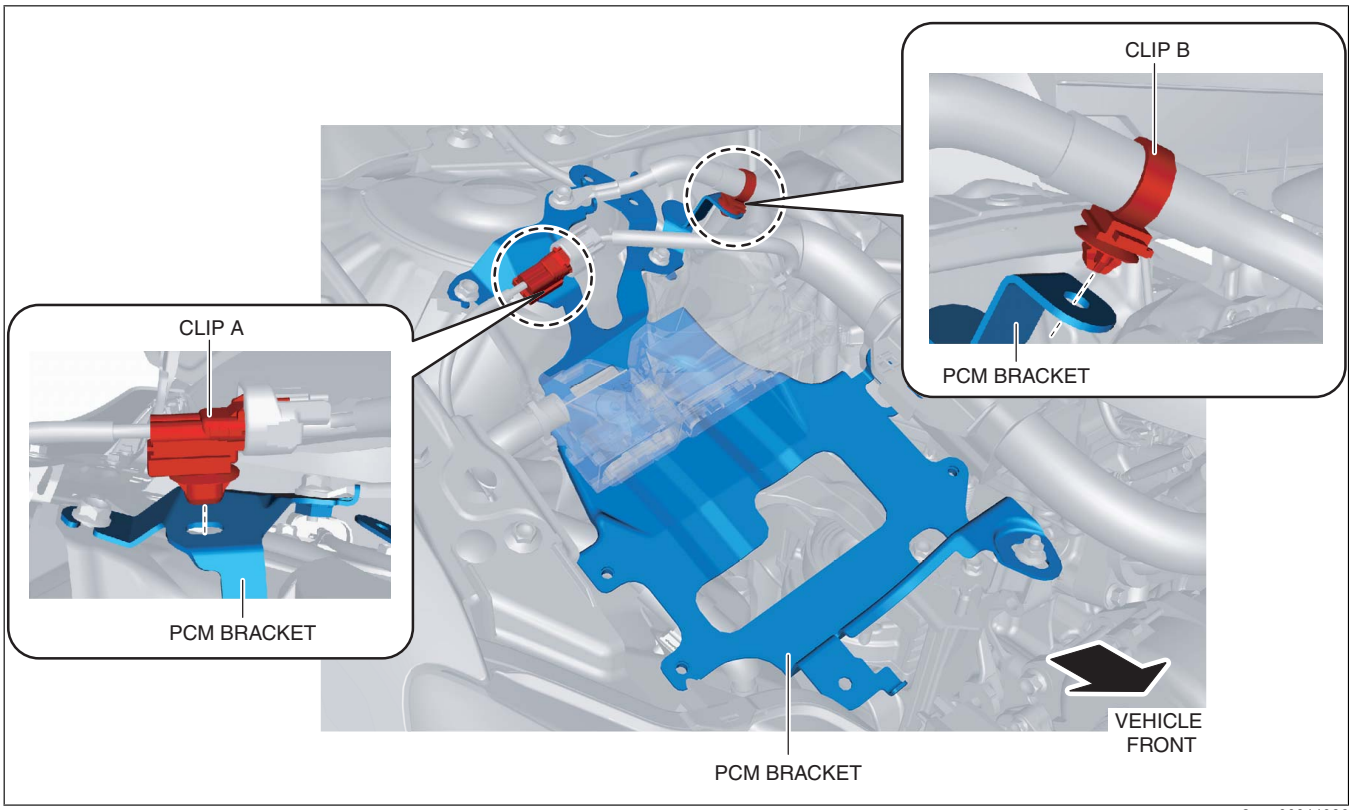

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5. Remove the bolt.
6. Remove the nut.
7. Remove the bracket.
8. Remove the PCM.



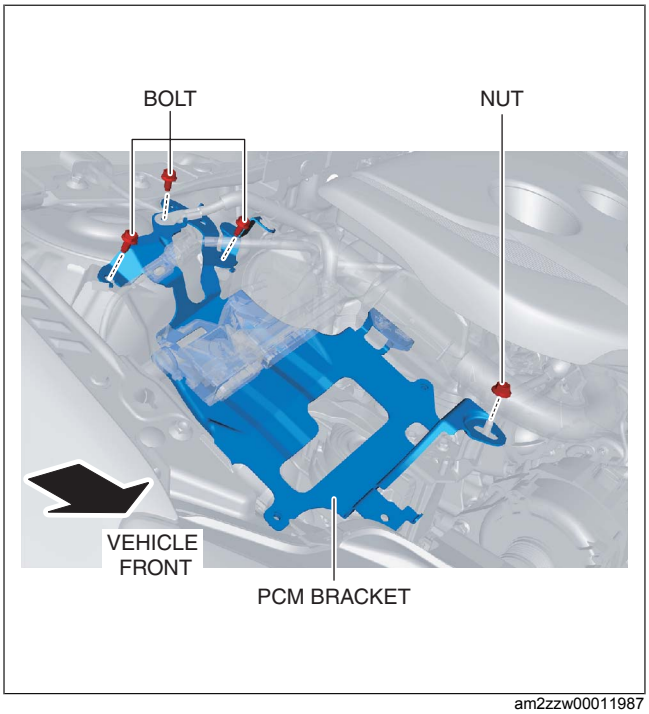

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9. Remove the clip A from the PCM bracket.




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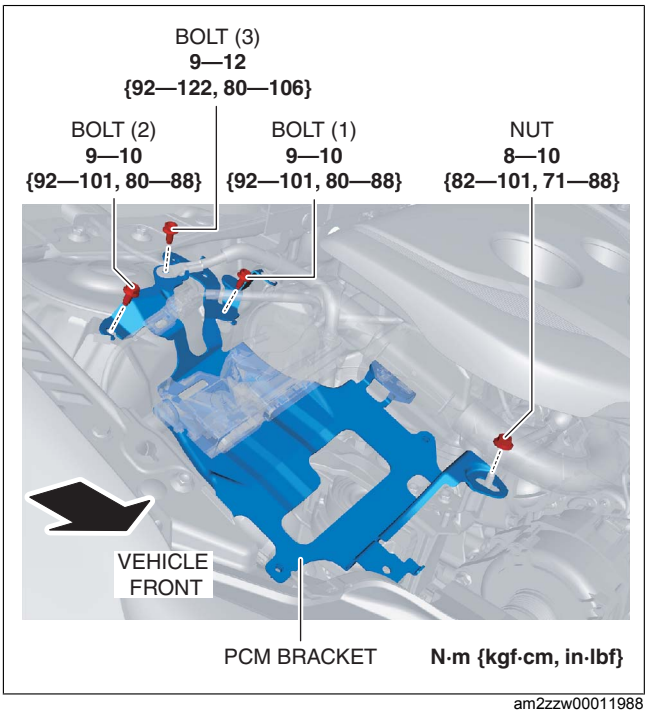
10. Remove the clip B from the PCM bracket.
11. Remove the nut.
12. Remove the bolt.
13. Remove the PCM bracket. (See PCM Bracket Installation Note.)
14. Install in the reverse order of removal.
15. When replacing the PCM on the vehicles, perform the “PCM Replacement”. (See PCM Replacement.)




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**PCM Bracket Installation Note**

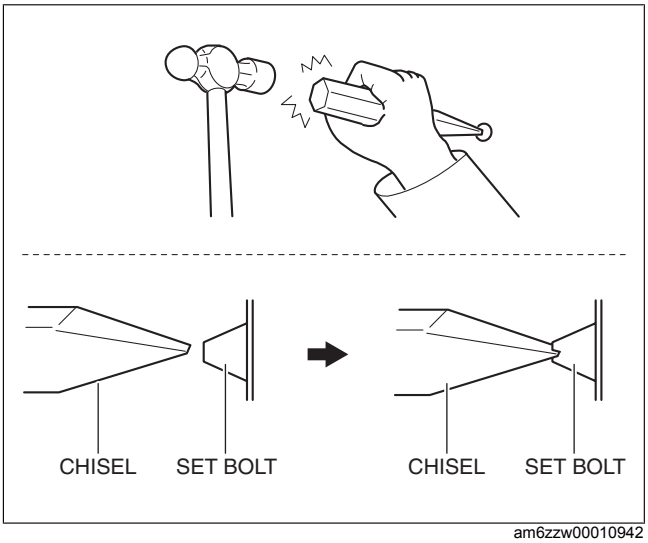
1. Install the PCM bracket using the following procedure:
   (1) Temporarily tighten the bolt (1).
   (2) Install the bolt (2).
   (3) Install the nut.
   (4) Tighten the bolt (1).
   (5) Install the bolt (3).



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**Set Bolt Removal Note**

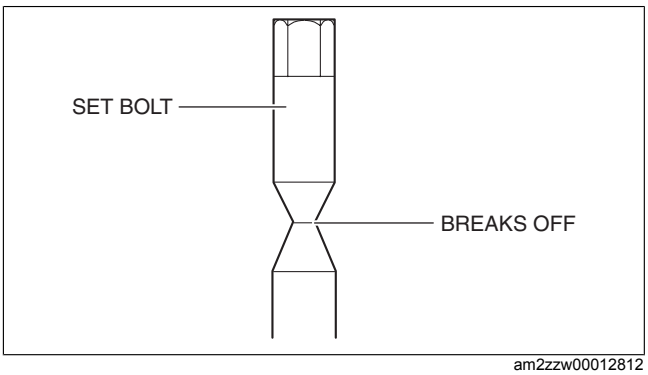
1. Using a chisel and a hammer, cut a groove on the head of the set bolt so that a screwdriver can be inserted.
2. Loose the set bolt using an impact screwdriver or pliers.



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**Set Bolt Installation Note**

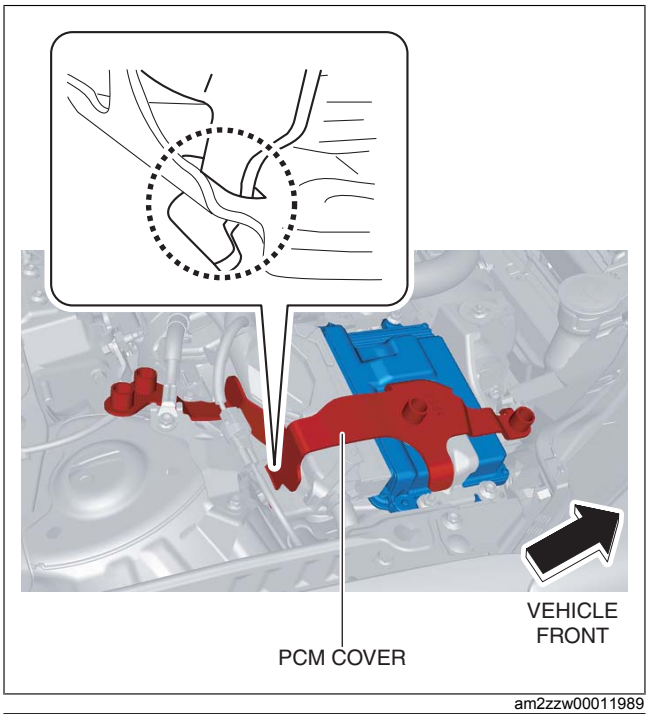
1. Install a new set bolt and tighten it until the neck of the bolt breaks off.



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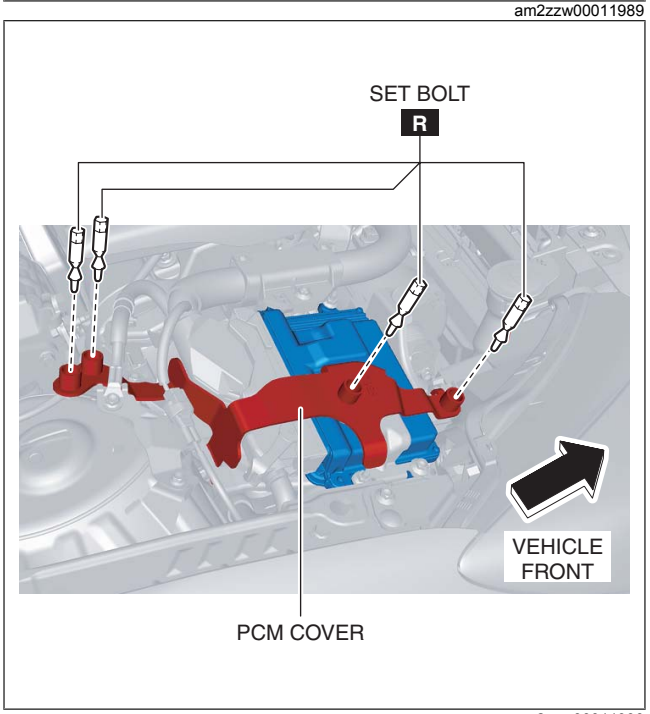
**PCM Cover Installation Note**

1. Install the PCM cover as shown in the figure.




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2. Temporarily tighten the four bolts, then completely tighten them.




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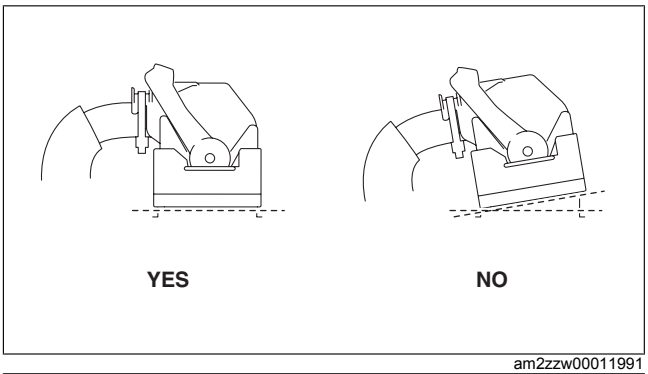
**PCM Connector Connection Note**

**Caution**

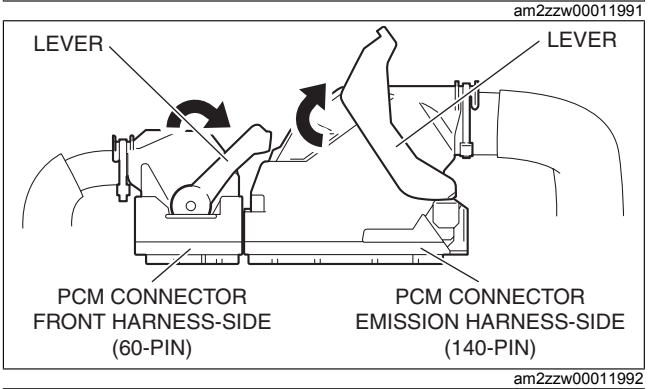
- **Do not touch the PCM connector terminal. The terminal is extremely thin and can be damaged by touching it.**

- **If the PCM connector is inserted at an angle and the lever is moved, the connector could be damaged. Verify that the PCM connector is inserted straight.**

1. Set the PCM connector lever to the position shown in the figure.

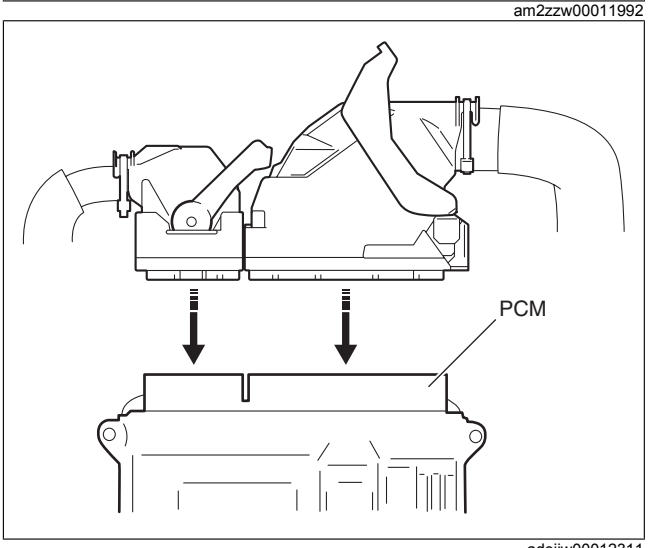



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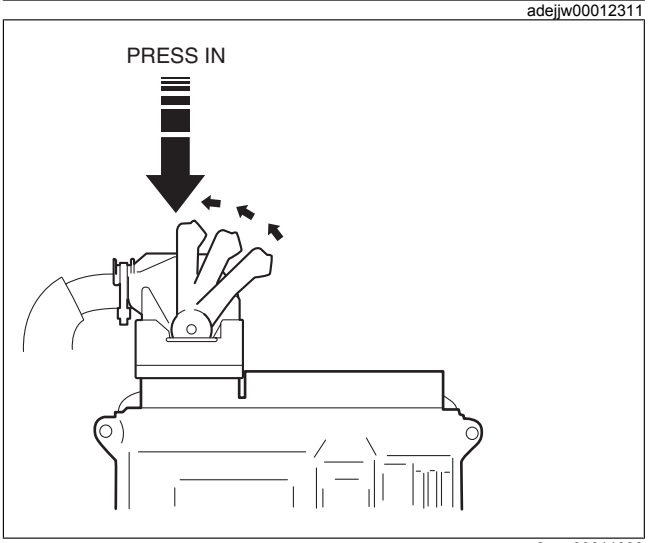

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2. Align the PCM connector straight against the connection surface.



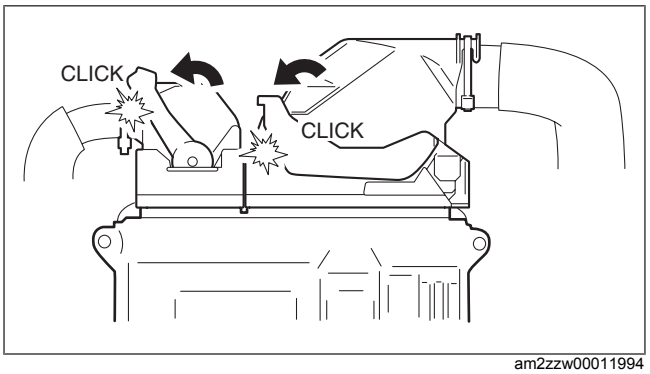

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3. Insert the PCM connector straight and press it in until the lever moves up naturally. (Front harness-side connector)




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4. Press the PCM connector lever until a click sound is heard.



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